



Notes for submissions

- Please pack your solution into a single zip-archive.
- If there is something non-obvious necessary for compiling or running the code, please add a README file.

Exercise 1: 2D Lennard-Jones Monte Carlo NVT Simulation

(20 points)

- (a) (2 points) Initialize N particles randomly and uniformly inside a square box of side length L , ensuring no two particles are closer than a small minimum distance r_0 .

Write a function `apply_pbc(x)` that maps any coordinate x back into $[0, L)$, and a function `minimum_image(xi, j)` that implements the minimum image convention between particle i and image particle j .

- (b) (2 points) Let us consider the shifted Lennard-Jones potential:

$$U_{\text{shift}}(r) = \begin{cases} 4\epsilon \left[\left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^6 \right] - U_{LJ}(r_c), & r < r_c \\ 0, & r \geq r_c \end{cases}$$

Implement a function that calculates the potential energy U_{shift} , and another function that computes the total potential energy U_{tot} by summing over all unique pairs $i < j$ using your minimum image and LJ routines. Use $\epsilon = 1$, $\sigma = 1$, $r_c = 2.5$.

- (c) (4 points) Select a random particle i and displace it by a vector (δ_x, δ_y) with each component in $[-\Delta, +\Delta]$, and recompute the energy change ΔE for the move by recalculating only the interactions of particle i with the other $N - 1$ particles.
- (d) (2 points) Implement the above proposal move as a function and include the Metropolis acceptance criterion

$$P_{\text{acc}} = \min \left(1, e^{-\beta \Delta E} \right), \quad \beta = \frac{1}{T}$$

into a single “sweep” of N attempted moves (a single attempt per particle) and track the acceptance ratio.

- (e) (4 points) Write a routine that equilibrates the system for n_{eq} sweeps, then performs n_{samp} sampling sweeps, recording the total energy every m sweeps. Then, compute the average energy per particle $\langle E \rangle / N$, its standard error, and discuss how the choice of Δ affects the acceptance ratio.
- (f) (6 points) We can compute the pressure P using the following expression

$$P = \frac{\rho}{\beta} + \frac{W}{A} + \Delta P_{\text{tail}}$$

where A corresponds to the box area, W to the virial

$$W = \frac{1}{2} \sum_i \sum_{i < j} \mathbf{f}_{ij} \cdot \mathbf{r}_{ij}$$

and ΔP_{tail} to the long-range correction

$$\Delta P_{\text{tail}} = \pi \rho^2 \epsilon \left[\frac{24}{11} \left(\frac{\sigma}{r_c} \right)^{12} - \frac{12}{5} \left(\frac{\sigma}{r_c} \right)^6 \right]$$

due to the potential energy at cutoff at r_c .

Plot the pressure P as a function of the density ρ , for temperature $T = 2$ and $\rho \in [0.01, 0.5]$. Use $\Delta = 0.65$.

The ideal gas law corresponds to $P(\rho) = \frac{\rho}{\beta}$. Plot it for the same density range and compare it with your results.